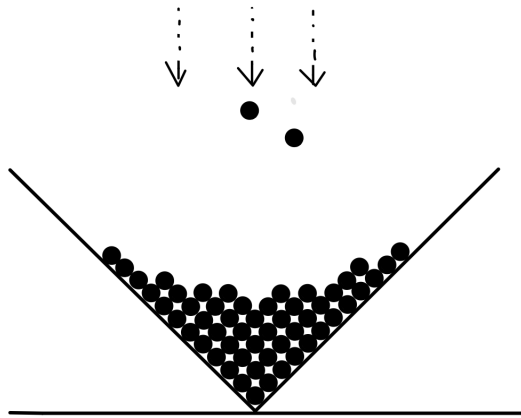


Model Solvay 2026

Problem: Two-Dimensional Crystal Growth in a Wedge

Identical circular crystals of radius r are released sequentially into a two-dimensional wedge-shaped region (see figure). The wedge has an opening angle of 90° , with each side inclined at 45° to the horizontal. Each crystal falls under gravity and, upon contacting either the wedge boundaries or previously deposited crystals, settles into a mechanically stable position. A crystal may occupy any available stable position; it is not required to reach the configuration of minimum gravitational potential energy, but it must be mechanically supported.



Assume that the probability of a deposition shape formed by N crystals is proportional to the square of the total number of distinct deposition sequences leading to that shape.

1. For $N = 2, 3, 4$, and 5 , list all possible stable deposition shapes and compute the probability of occurrence of each shape.
2. Describe a suitable way to specify an arbitrary stable deposition shape formed by N identical crystals in the wedge-shaped region.
3. Using the combinatorial results obtained for small N in Part (1), derive a general expression for the probability of obtaining a given deposition shape when any N crystals are deposited sequentially.
4. Using this expression, determine the condition for the most probable deposition shape for a fixed N number of crystals.

5. Now assume the crystals are infinitesimally small, with radius $r \rightarrow 0$, while the total deposited area remains fixed,

$$N\pi r^2 = A.$$

In the continuum limit $N \rightarrow \infty$ and $r \rightarrow 0$, keeping A fixed, determine the resulting shape of the boundary of the most probable deposited region.

6. Propose and describe a numerical simulation that models the sequential deposition of crystals in the wedge and is capable of extracting the asymptotic shape of the deposited region for large N .